

PAPER 1

JULY 2014 MID-YEAR SESSION

2 Hours

Additional materials:
Answer Paper
Data Booklet
Electronic calculator

Time: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name in the space on top of this paper and on all separate answer sheets used.

Answer **two** questions from section A, **one** from section B, **two** from section C and **one** other question chosen from any section.

Answer **six** questions

Section A

Write your answers on separate answer sheets provided.

At the end of the examination, fasten any separate answer paper used securely to the question paper

All essential working must be shown.

INFORMATION

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.



Peace all the Time....

MARK

- (a) Fig. 1 shows the variation in the first ionisation energies of the elements Na to Ar.

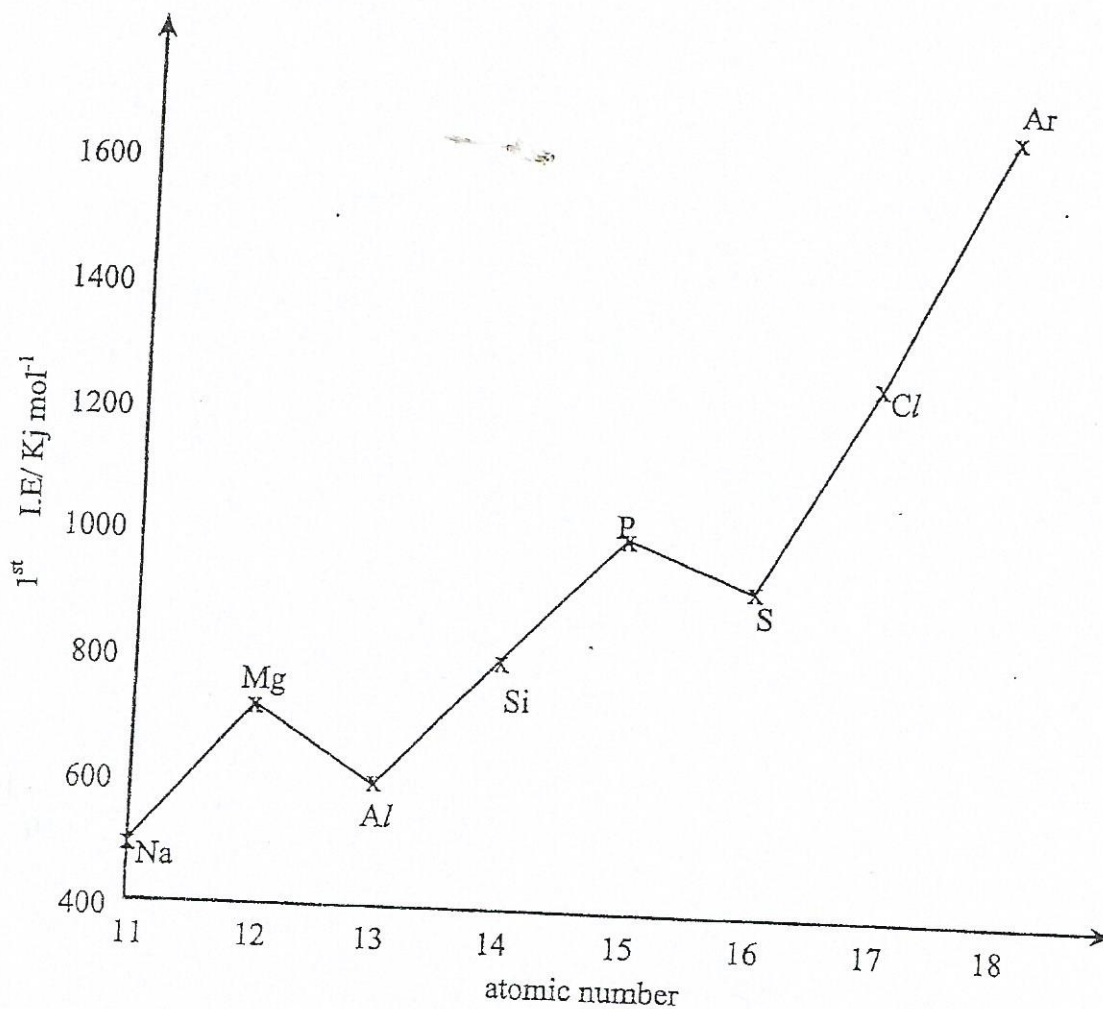


Fig. 1

- (i) Define the term *ionisation energy*.

(ii) Explain why the first ionisation energy

1. generally increases across the period,

2. of magnesium is greater than that of aluminium,

3. of phosphorous is greater than that of sulphur.

(iii) Identify an element from Na to Ar that has the highest melting point. Explain your answer.

element _____

reason(s) _____

[9]

- (b) The first ionisation energies of magnesium and calcium are 736 kJmol^{-1} and 590 kJmol^{-1} respectively.

(i) Write an equation which represents the first ionisation of magnesium

(ii) Explain why the first ionisation energy of calcium is lower than that of magnesium.

[2]

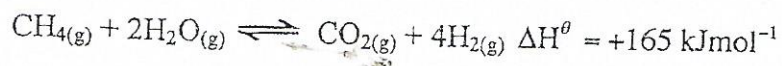
- (c) First ionisation energy shows periodicity.

Explain the term *periodicity*.

[1]

[Total: 12]

- 2 Hydrogen can be obtained from methane by the reaction shown below.



- (a) Initially, 1.5 moles of methane and 2.0 moles of steam were placed in a 5.0 dm^3 flask and heated in the presence of a catalyst until equilibrium was established. The equilibrium mixture contained 0.5 moles of carbon dioxide.

- (i) Write an expression for the equilibrium constant, K_c , for this reaction.
-

- (ii) Calculate the concentrations of methane, steam and hydrogen in the equilibrium mixture.

methane

steam

hydrogen

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- (iii) Calculate K_c at this temperature and give its units.

$K_c =$ _____ [6]

- (b) State and explain the effect, on the amount of hydrogen produced of

- (i) increasing the volume of the flask to 8.0 dm^3 ,

- (ii) reducing the temperature.

[4]

[Total: 10]

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- 4 (a) Give the formulae of the organic product(s) formed when 3-hydroxypropanal, $\text{HO}(\text{CH}_2)_2\text{CHO}$, reacts with

(i) Tollen's reagent,

(ii) acidified potassium dichromate,

(iii) NaBH_4 .

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[3]

- (b) 3-bromopentane, can undergo reactions 1, 2 and 3 as shown in Fig 2.

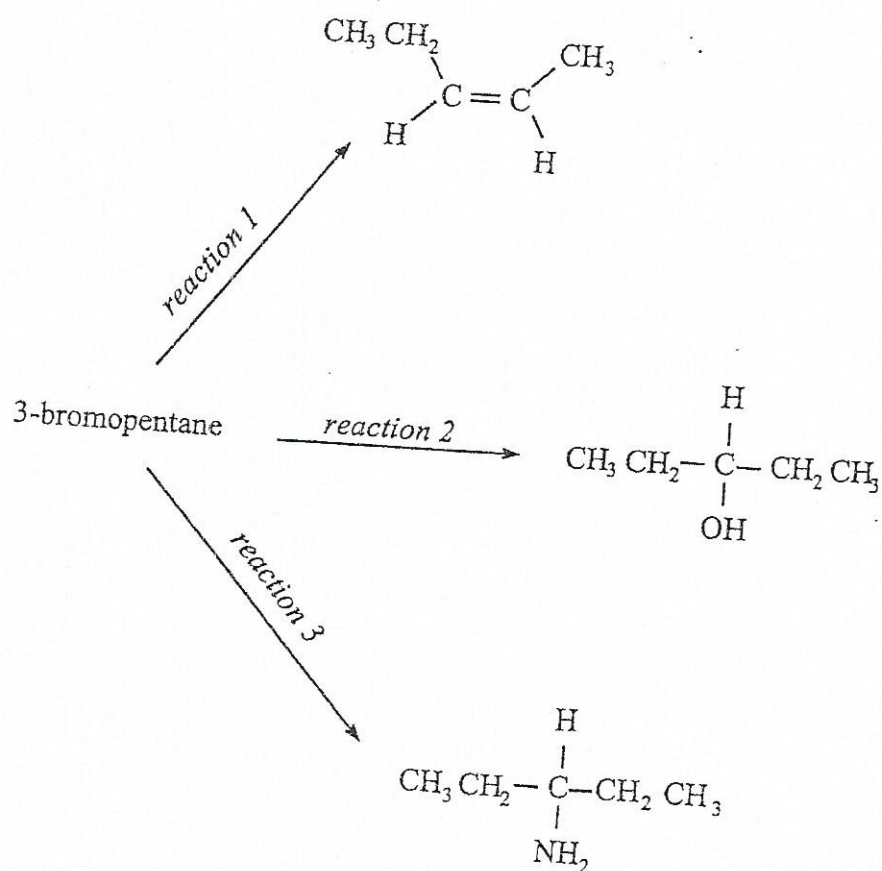


Fig. 2

- (i) Draw the displayed formula of an isomer of 3-bromopentane that has a chiral centre.

- (ii) State the type of reaction undergone, the reagent(s) and the conditions used for reactions 1 and 3.

reaction 1

Type of reaction _____

Reagent _____

Condition(s) _____

reaction 3

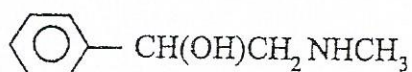
Type of reaction _____

Reagent _____

Condition(s) _____

[7]
[Total: 10]

- 5 (a) Fig. 3 shows the structure of adrenaline.



adrenaline

Fig. 3

- (i) State the type of isomerism found in the adrenaline molecule.

- (ii) Draw the displayed structural formulae of the isomers of adrenaline.

[3]

- (iii) Draw the displayed formula of the organic product formed when adrenaline is reacted with acidified potassium dichromate.

[1]

- (b) Fig. 4 shows the displayed structural formulae of procaine.

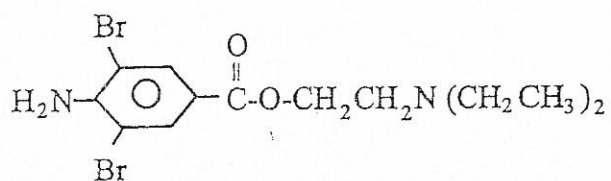


Fig. 4

Give the organic product(s) formed when procaine reacts with

- (i) NaOH ,

(ii) $\text{Br}_{2(\text{aq})}$,

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(iii) Nitrous acid + HCl below 10°C followed by phenol.

[4]
[Total: 8]